



Vidyavardhini's College of Engineering & Technology

Department of Civil Engineering

Course Name: Building Materials and Concrete Technology (2093113) **Sem/Year:** III/SE
Academic Year: 2026-27

Assignment No.: 01

Date of Submission: 14/08/2026

Marks: 10

2093113.1: Students will be able to identify suitable building materials based on properties, manufacturing, and construction use.

Q.No	Questions	BL
Q.1)	A commercial building is being constructed in a hot climate. Select the most suitable type of glass for the façade and justify your selection based on thermal performance, energy efficiency, and intended application.	3
Q.2)	For the construction of a two-storey residential building, select suitable construction materials for the foundation, walls, roof, flooring, and doors/windows, giving reasons for each selection.	3
Q.3)	A timber supplier provides both air-seasoned and kiln-seasoned timber. Select the appropriate type for door and window frames and justify your choice based on construction requirements.	3
Q.4)	A construction project requires large granite blocks for a bridge foundation. Select a suitable quarrying method and justify your choice based on the type of stone and site conditions.	3
Q.5)	A hospital project requires fire-resistant, lightweight, and thermally efficient walling materials. Select the most suitable material from refractory bricks, AAC blocks, and burnt clay bricks, and justify your recommendation.	3
Q.6)	Using a labelled cross-section of a tree trunk, identify the parts responsible for strength, growth, and durability, and explain their importance in selecting timber for construction.	3
Q.7)	Draw a labelled sketch of a modular burnt clay brick with standard dimensions and demonstrate how its dimensions help in planning brick masonry work.	3

Signature of Subject Incharge